(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: BS-109 ***Subject: Environmental Studies***

Time: 3 Hours ***Maximum Marks: 60***

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit.

Q1 Attempt all questions:-
(a) Discuss the values of biodiversity. **(2)**
(b) Discuss the urban problems related to energy. **(2)**
(c) Explain the importance of women and child welfare program. **(2)**
(d) Discuss the causes of soil pollution and their control. **(2)**
(e) Explain the term 'Population explosion'. Enumerate its effects. **(2)**
(f) Briefly discuss the current environmental issues. **(2)**

UNIT-I

Q2 (a) Discuss the establishment of an Ecosystem. What do you understand by the term Ecological Succession? **(4)**
(b) Why biogeochemical cycles are important for an Ecosystem. Make a well labeled diagram of any one cycle. **(4)**
(c) Write short note on: (i) Biopiracy (ii) Impact of deforestation **(4)**

Q3 (a) What is meant by biodiversity? Discuss the ways to conserve biodiversity. **(4)**
(b) What are the three forms of ocean energy? Explain the process of OTEC in brief. **(4)**
(c) Write short note on: (i) Ecological pyramids (ii) Biodiversity hotspots **(4)**

UNIT-II

Q4 (a) Enumerate and explain the various methods to air pollution control with suitable diagram. **(6)**
(b) (i) What is 'Thermal pollution'? State and explain the various sources of thermal pollution.
(ii) 'Recycling is an integral part of solid-waste management'. Comment. **(6)**

Q5 (a) Name and explain, the various ways of secondary treatment of waste water treatment. **(6)**
(b) (i) How you will define 'Disaster management'? Briefly describe the role of disaster manager and importance of advance planning in disaster management. **(4)**
(ii) Write short note on Incineration. **(2)**

UNIT-III

Q6 (a) What do you mean by sustainable development? Discuss the concept of sustainable development. **(4)**
(b) What are greenhouse gases? Discuss their contribution to global warming. **(3)**
(c) Write short note on:- (i) Acid rain (ii) Rainwater harvesting. **(5)**

Q7 (a) Briefly discuss the objectives of Resettlement and Rehabilitation Policy. **(4)**
(b) What are wastelands? Name and discuss the various methods of wasteland reclamation. **(3)**
(c) Describe briefly the provisions of the following acts (i) Air (prevention and control of pollution) Act, 1981 (ii) Water (prevention and control of pollution) Act, 1984. **(5)**

UNIT-IV

Q8 (a) Briefly discuss the problems of 'Family welfare programme'. **(3)**
(b) What is 'value education'? Discuss the concept of value education with suitable example. **(3)**
(c) What do you understand by 'Human Rights'? List the important Articles of the 'Declaration of Human Rights, 1948' adopted and proclaimed by United Nations. **(3)**
(d) Briefly describe the various schemes launched for women and child welfare in India. **(3)**

Q9 (a) Define the following terms (i) Population density (ii) Fundamental rights **(6)**
(b) What is AIDS? What are the sources and mode of transmission of HIV infection? **(3)**
(c) Comment on the role of IT in environment protection and human health. **(3)**

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